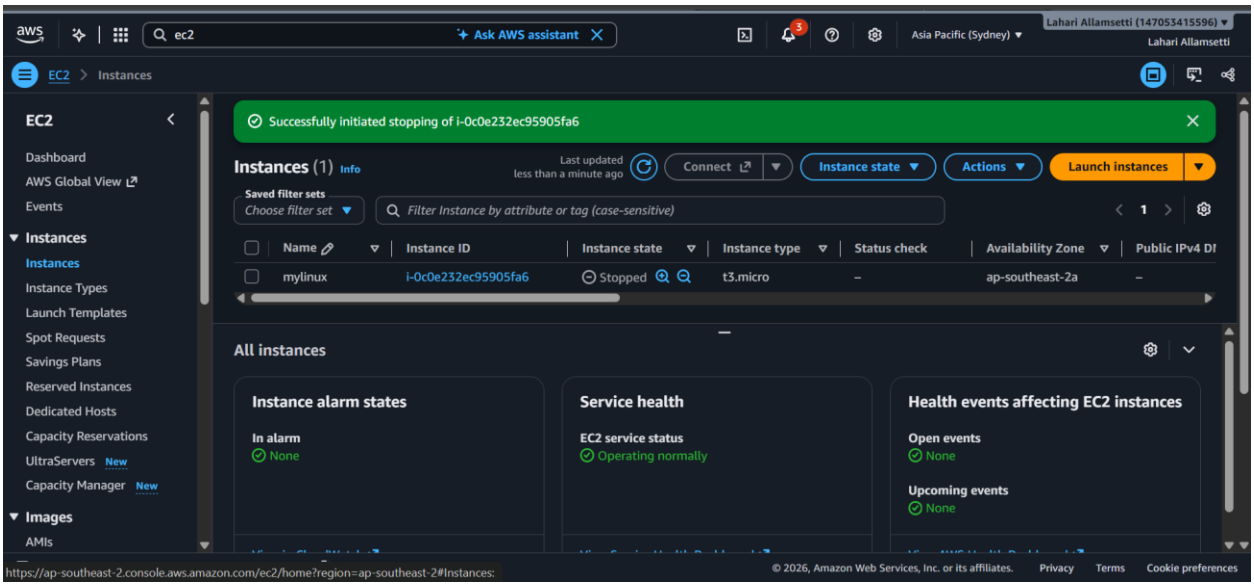
WEEK-2: CREATING A LINUX VIRTUAL MACHINE USING AWS EC2

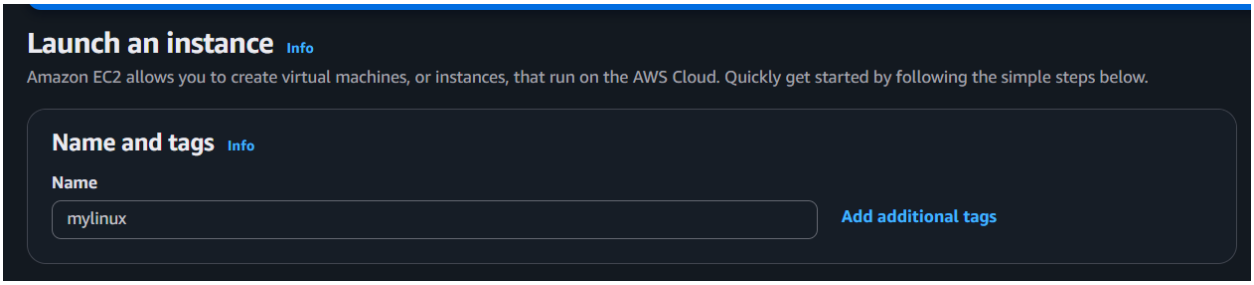
EC2 (Elastic Compute Cloud) is a service provided by Amazon Web Services (AWS) that allows users to create and manage virtual machines (virtual servers) in the cloud.

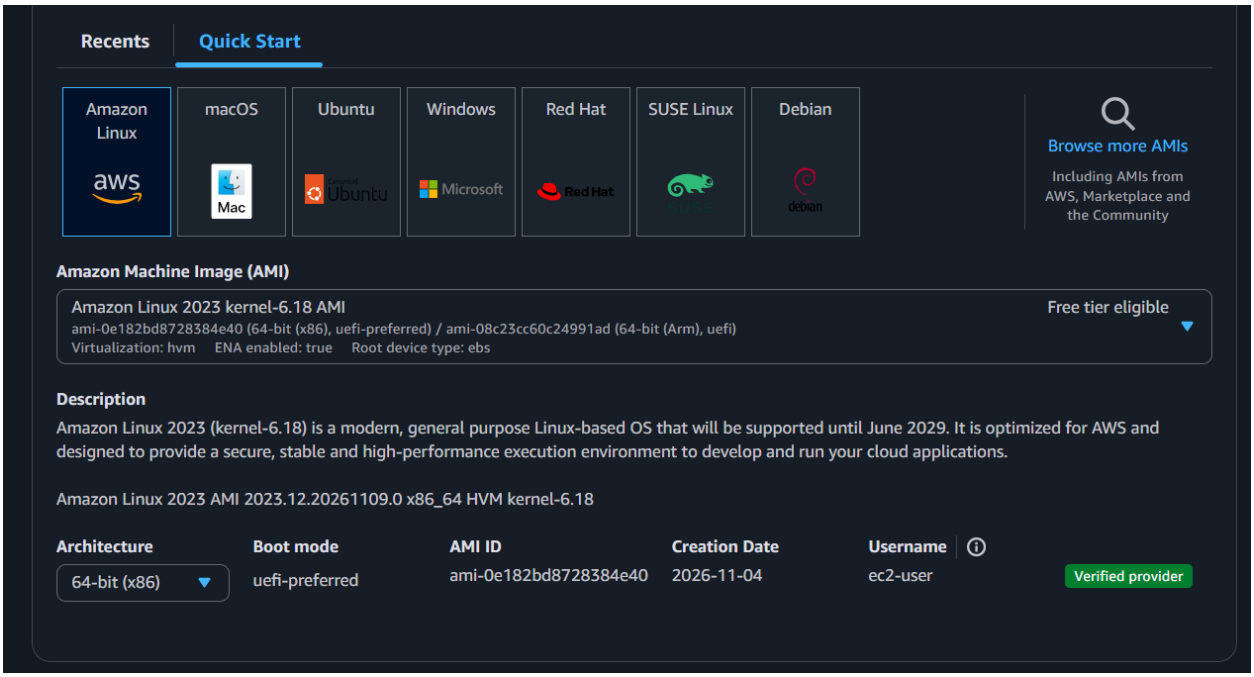
A Virtual Machine (VM) is a software-based computer that provides resources such as:

- CPU (Central Processing Unit)
- RAM (Memory)
- Storage
- Network connectivity
- Operating System

we create a Linux virtual machine using Amazon EC2.







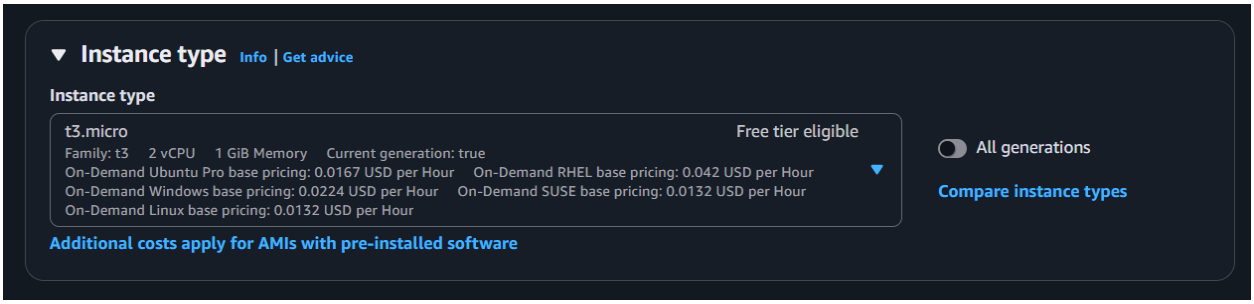
2. Selecting an Amazon Machine Image (AMI)

AMI (Amazon Machine Image) is a template used to create an EC2 instance. It contains the operating system and the necessary configuration required to launch a virtual machine.

We selected **Amazon Linux 2023** as the operating system for our virtual machine.

Some of the operating systems available in AWS include:

- Amazon Linux
- Ubuntu
- Windows
- Red Hat Linux
- SUSE Linux
- Debian



Instance Types

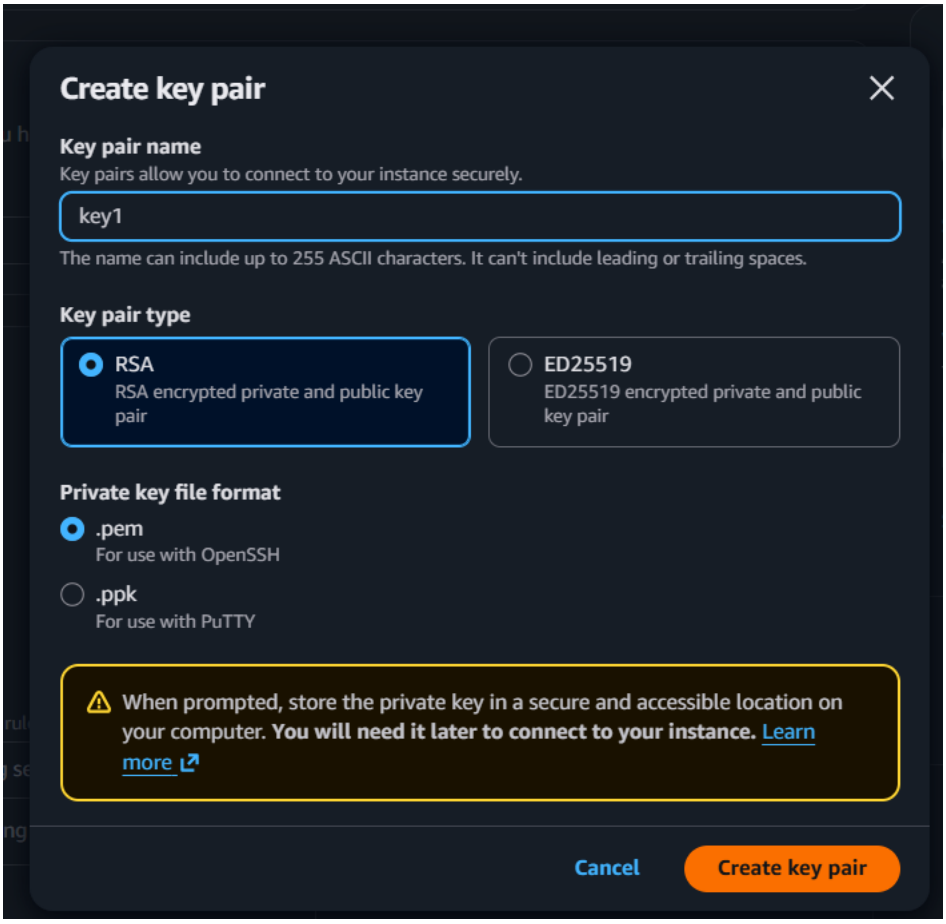
An instance type determines the computing resources allocated to the virtual machine, such as CPU and memory.

The following instance types were discussed:

- **t2.micro:** A small, burstable general-purpose instance.
- **t3.micro:** A newer burstable general-purpose instance.

In this practical, we selected the **t3.micro** instance type.

It provides basic computing resources required to run applications and perform Linux operations.



3. Creating a Key Pair

A **Key Pair** is used for secure authentication when connecting to an EC2 instance.

It consists of two keys:

- **Public Key:** Stored on the EC2 instance.
- **Private Key:** Downloaded and securely stored on our local computer.

Steps to create a key pair:

1. Enter a name for the key pair.
2. Select the RSA algorithm.
3. Choose the private key file format as .pem.
4. Click on Create Key Pair.
5. Download and save the private key securely.

   RSA Algorithm

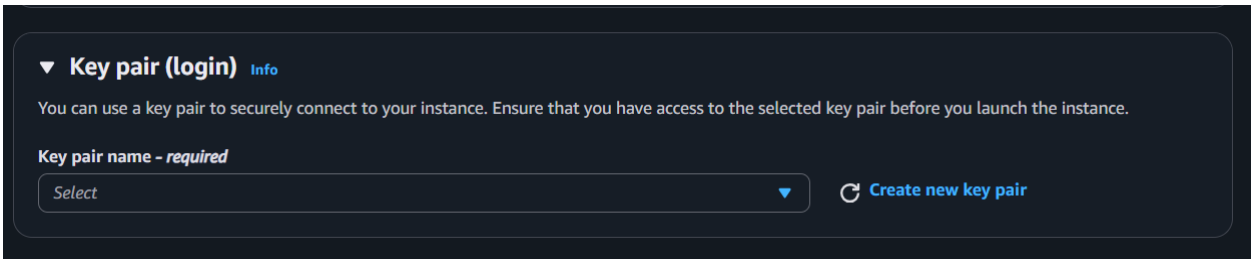
RSA (Rivest–Shamir–Adleman) is a public-key cryptographic algorithm used for secure communication and authentication.

It uses two keys: a public key and a private key.

Encryption and Decryption

- **Encryption:** The process of converting readable data into an encoded format to protect it.
- **Decryption:** The process of converting encrypted data back into its original readable format.

These techniques help protect information during communication.



4. SSH (Secure Shell)

SSH stands for **Secure Shell**. It is a network protocol used to securely connect to and manage a remote computer or server.

In this practical, SSH is used to connect to the Linux EC2 instance from our local computer.

SSH provides:

- Secure remote login.
- Encrypted communication.
- Secure execution of commands on a remote machine.
- Authentication using an SSH key pair.

Note: SSH is used to connect to an already-created server. The EC2 instance itself is created using the AWS Launch Instance option.

SSH commonly uses port 22, whereas HTTPS commonly uses port 443. Both are different protocols.

5. Launching the EC2 Instance

After selecting the AMI, instance type, and key pair, we review the instance configuration.

The launch summary includes:

- Number of instances: 1
- Operating System: Amazon Linux 2023
- Instance Type: t3.micro
- Security Group: New security group
- Storage: 8 GiB

After verifying the details:

1. Click on **Launch Instance** on the right side of the screen.
2. AWS starts creating the virtual machine.
3. Wait until the instance is successfully launched.
4. Open the EC2 Instances page.
5. Verify that the instance state changes to **Running**.

6. Connecting to the EC2 Instance

After successfully launching the instance:

1. Open the EC2 Instances page.
2. Locate the created Linux instance.
3. Select the checkbox beside the instance.
4. Click on the **Connect** button.
5. Choose an available connection method, such as EC2 Instance Connect or SSH.
6. Complete the connection process.

After connecting successfully, we can access the Linux terminal and execute Linux commands on the remote EC2 instance.

7. EC2 Status Checks

AWS performs status checks to verify whether the EC2 instance and its supporting resources are functioning correctly.

The checks can include:

When the console displays **3/3 checks passed**, it indicates that all three displayed checks have passed successfully.

E. Displaying File Content Using the CAT Command

The cat command is used to display the contents of a file in the terminal.

Syntax:

cat filename

Example:

cat notes.txt

This displays the content saved inside the notes.txt file.

F. Listing Files and Directories

The ls command displays the files and directories present in the current directory.

Example:

ls

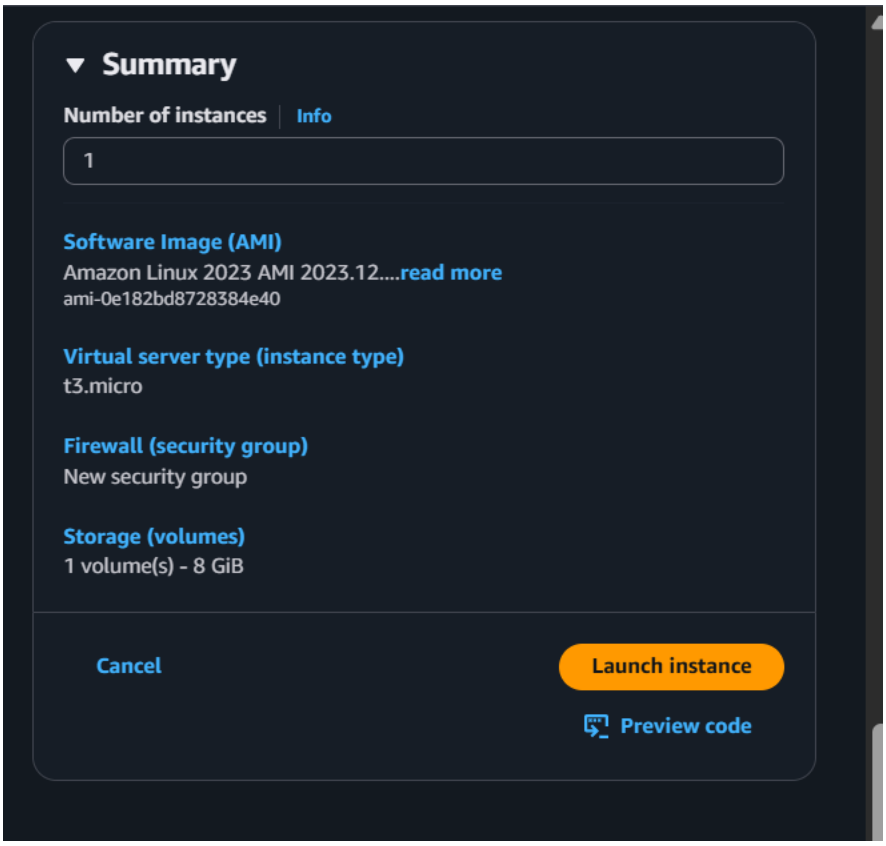
G. Checking the Current Directory

The pwd command displays the present working directory.

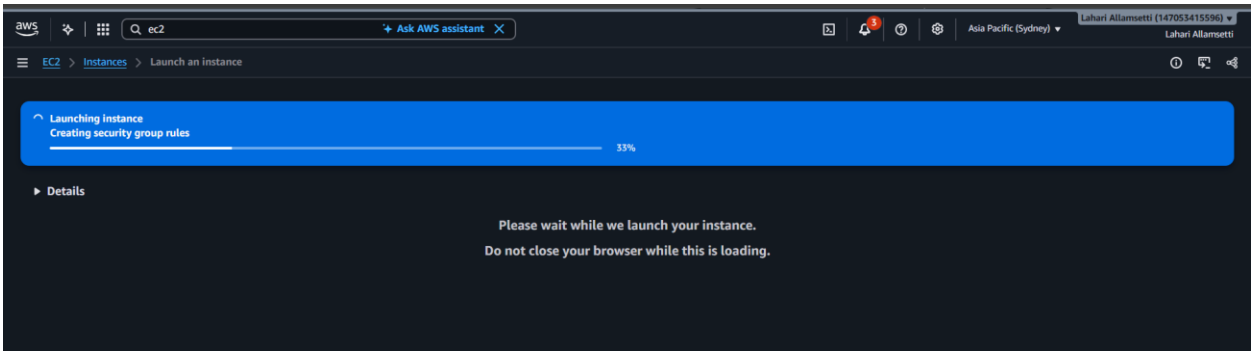
Example:

pwd

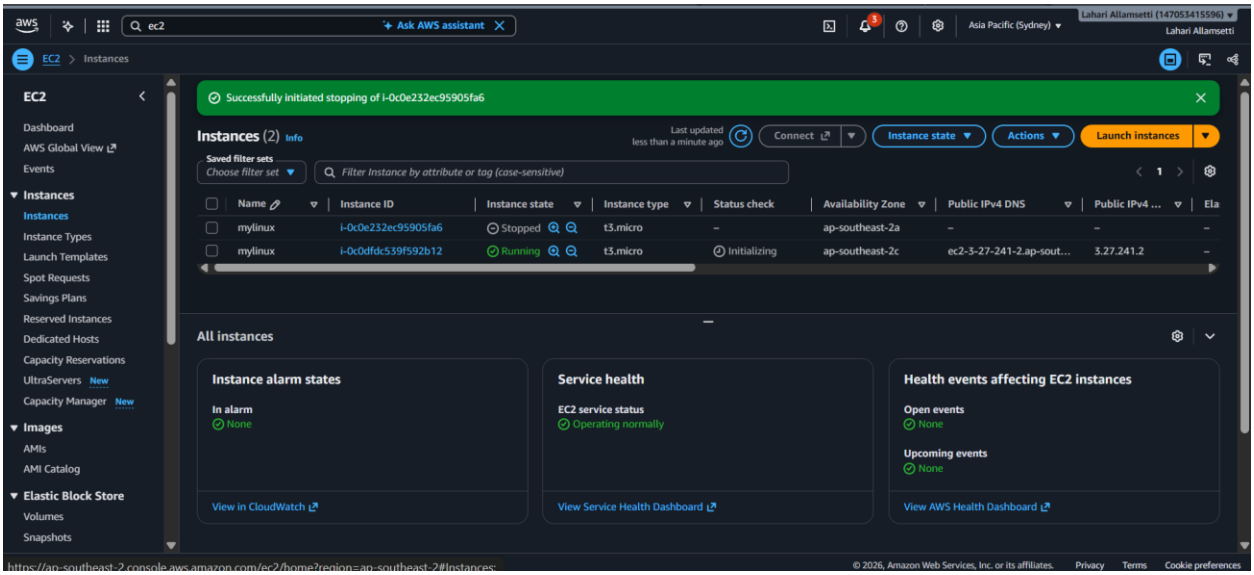
Successfully created and launched a Linux virtual machine using Amazon EC2. Selected Amazon Linux 2023 as the operating system and t3.micro as the instance type. Created an RSA key pair for secure authentication, connected to the EC2 instance, and performed basic Linux operations such as creating directories, creating files, writing content, and displaying file contents using Linux commands.



Then click on launch instance on the right corner







8. Performing Basic Linux Operations

After connecting to the Linux instance, we perform basic file and directory operations using Linux commands.

A. Creating a Directory

The mkdir command is used to create a new directory (folder).

Syntax:

mkdir directory_name

Example:

mkdir CIS_practical

This creates a directory named CIS_practical.

B. Changing the Directory

The cd command is used to move from one directory to another.

Syntax:

cd directory_name

Example:

cd CIS_practical

This moves us into the newly created directory.

C. Creating a File

The touch command is used to create an empty file.

Example:

touch notes.txt

This creates a file named notes.txt.

D. Writing Content into a File

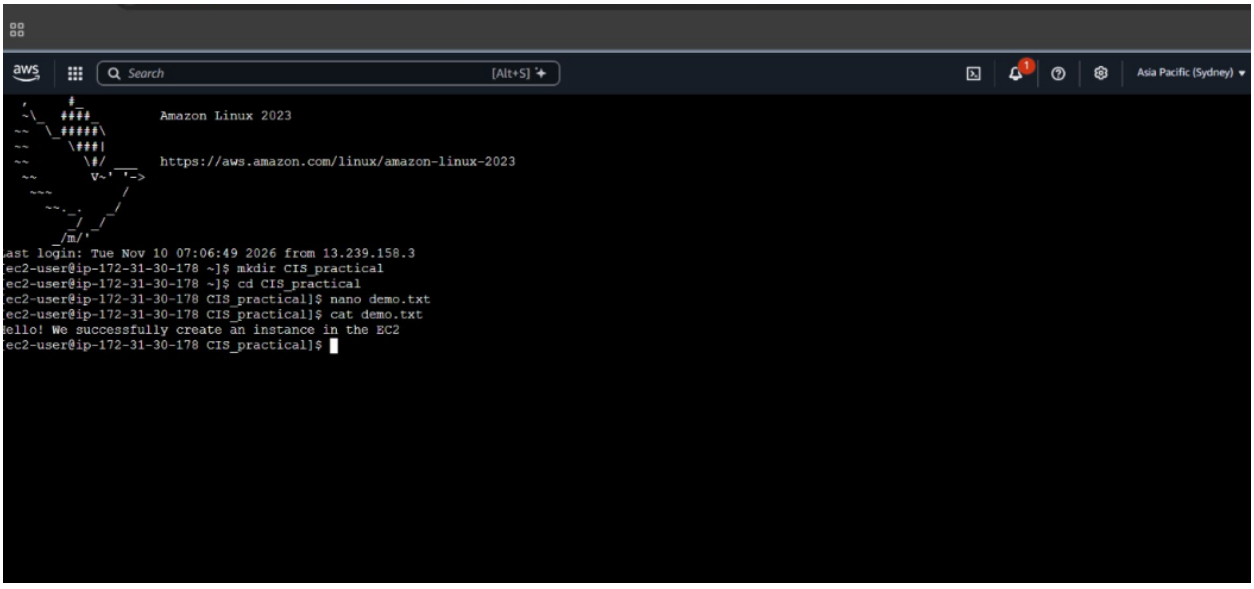
The echo command is used to display text. We can use the redirection operator (>) to save that text into a file.

Example:

echo "Welcome to Linux EC2" > notes.txt

This creates the file if it does not exist and saves the given content inside it. If the file already contains data, the existing content will be overwritten.

To append additional content without deleting the existing content, we use >>.



Successfully created and launched a Linux virtual machine using Amazon EC2. Selected Amazon Linux 2023 as the operating system and t3.micro as the instance type. Created an RSA key pair for secure authentication, connected to the EC2 instance, and performed basic Linux operations such as creating directories, creating files, writing content, and displaying file contents using Linux commands.